



 Production Architectural Enhancements (v2.0 Fixes)

PATCH 01

Concurrency Lock

Prevents race conditions when multiple operators inspect the same pulsing H3 cell. Firestore applies an optimistic lock with a 60-second TTL.

PATCH 02

Stale Telemetry Fallback

If weather or transit APIs timeout (>5m), the system freezes last-known lens scores and flags the H3 cell in a DEGRADED_STATE.

PATCH 03

15-Min Escalation Loop

Auto-monitors dispatched units. If status does not transition to 'In Transit' within 15 mins, an admin alert triggers automatically.

PATCH 04

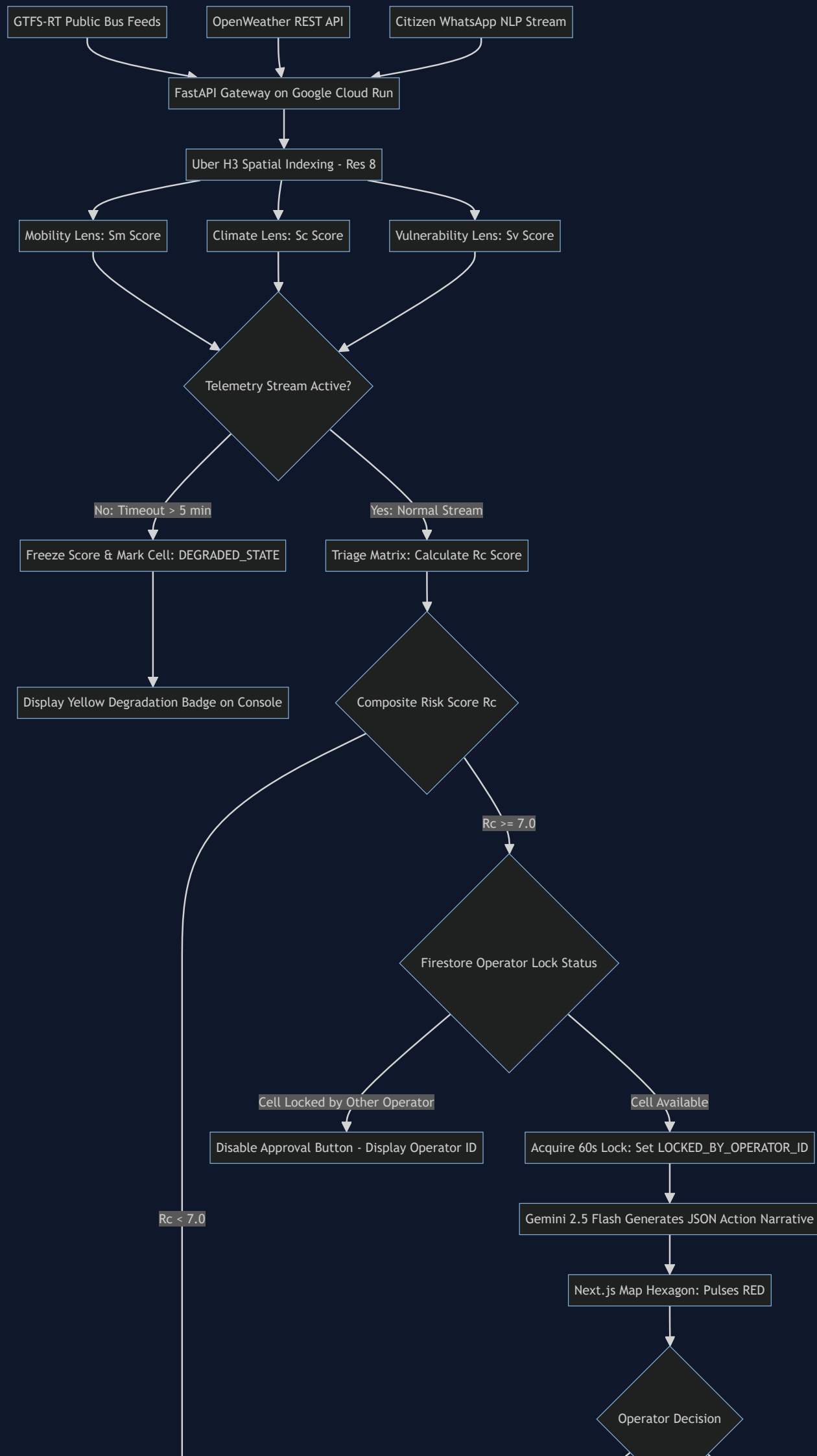
PWA Offline Outbox

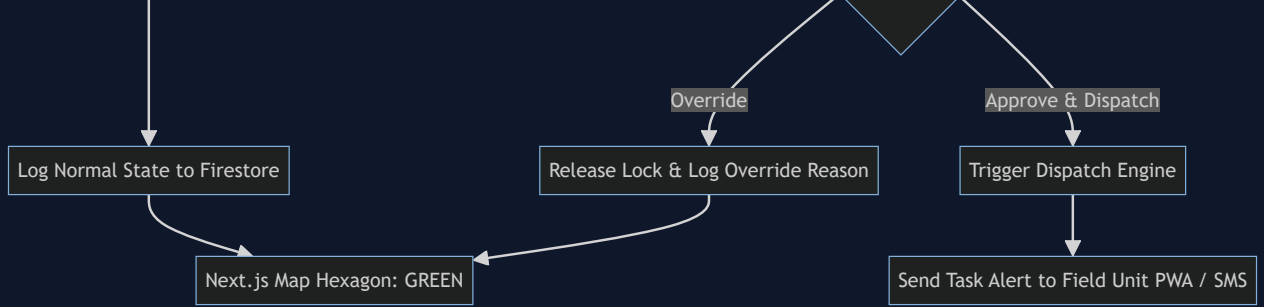
Field workers in low-connectivity areas store completion status and photos in IndexedDB, auto-syncing when network returns[cite: 2].



Flow Path 1: Automated Ingestion, Triage & Concurrency-Safe Dispatch[cite: 2]

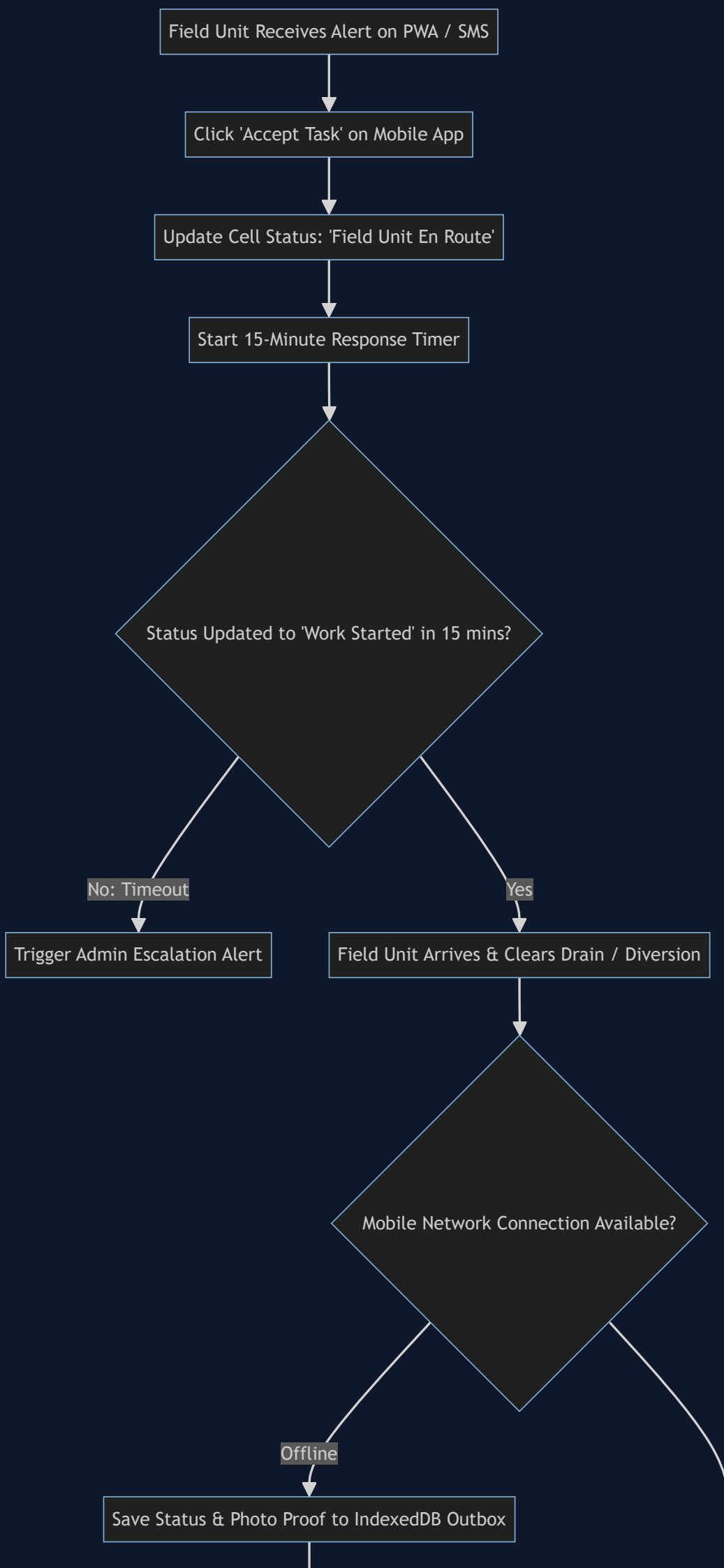
Includes telemetry health checks, algebraic scoring (\$R_c\$), Firestore optimistic locks, and Gemini 2.5 narration[cite: 1, 2].

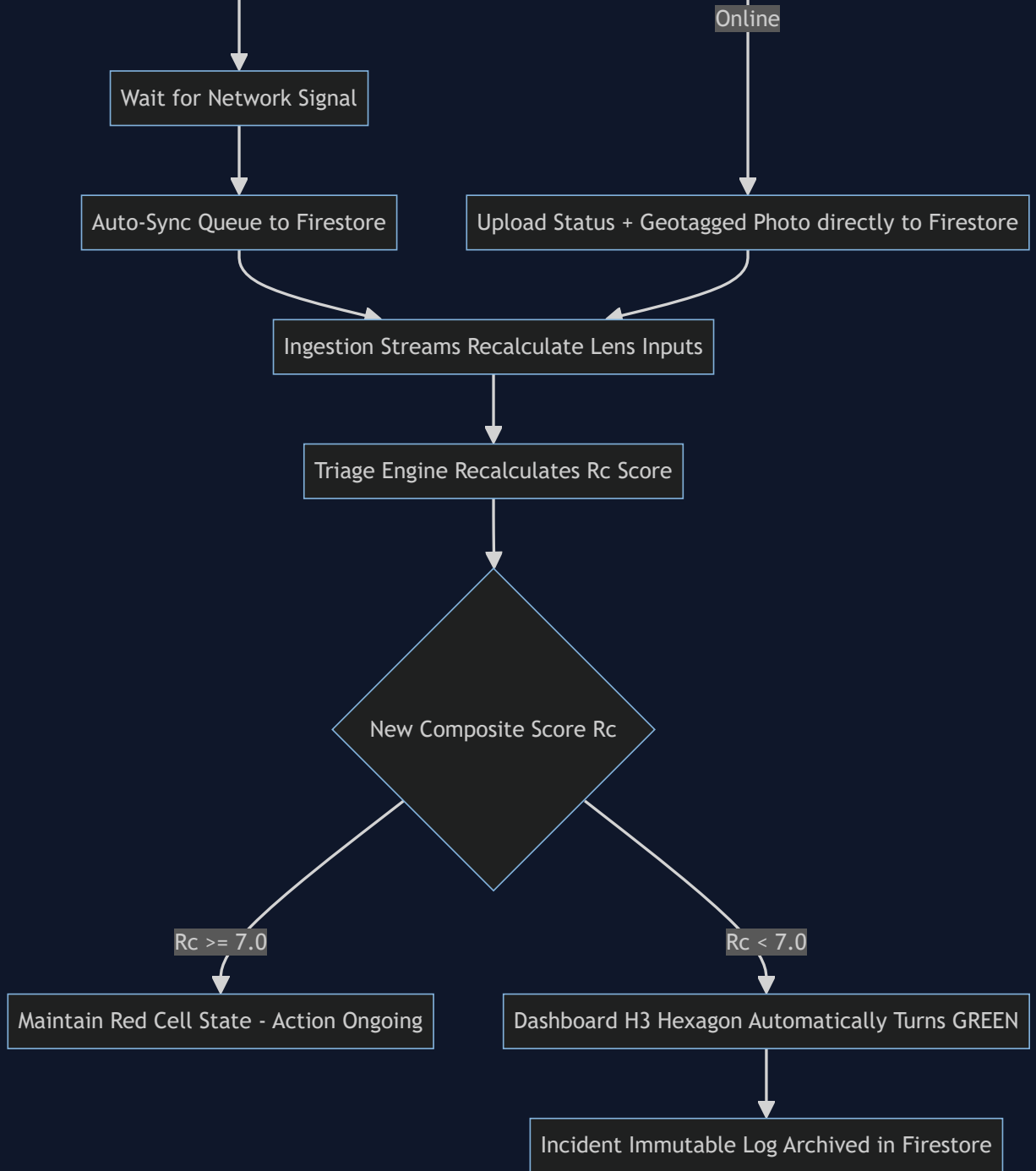




Flow Path 2: Field Specialist Action & Automated State Recovery[cite: 2]

Handles task acceptance, offline IndexedDB sync, physical clearance, and closed-loop grid recovery[cite: 2].





Application Screen Catalog & User Journey Specification[cite: 2]

Detailed breakdowns of user access levels, UI/UX core components, and key operations across the platform[cite: 2].

ID[cite: 2]	SCREEN NAME[cite: 2]	ACCESS LEVEL[cite: 2]	PRIMARY OBJECTIVE[cite: 2]	UI / UX COMPONENTS[cite: 2]	KEY FUNCTIONAL ACTIONS & PATCHES[cite: 2]
SCR-01	Authentication & Onboarding	Public / Role-Based	Secure multi-tenant city selection and entry point[cite: 2].	Glassmorphism panel, city dropdown, MFA input, role badge[cite: 2].	OAuth/Firebase Auth, select target jurisdiction (e.g., Sehore, Ashta)[cite: 1, 2].
SCR-02	Operator Main Console	Dispatch Operator	Real-time geospatial telemetry monitoring via H3 grid[cite: 2].	Mapbox WebGL map, Uber H3 Hex overlay, Lens toggle drawer[cite: 2].	Inspect H3 cells, toggle Mobility/Climate lenses, detect DEGRADED states[cite: 2].

ID[CITE: 2]	SCREEN NAME[CITE: 2]	ACCESS LEVEL[CITE: 2]	PRIMARY OBJECTIVE[CITE: 2]	UI / UX COMPONENTS[CITE: 2]	KEY FUNCTIONAL ACTIONS & PATCHES[CITE: 2]
SCR-03	H3 Detail & Action Drawer	Dispatch Operator	Review raw lens metrics and AI-synthesized dispatches[cite: 2].	Radar charts (\$S_m, S_c, S_v\$), Gemini 2.5 Flash narrative card[cite: 1, 2].	Acquire Firestore lock, approve/override dispatch, view active operator lock status[cite: 2].
SCR-04	Field Unit Mobile App	Field Specialist	Low-bandwidth task execution interface[cite: 2].	Vertical task card, Mapbox navigation, camera upload trigger[cite: 2].	Push/SMS alerts, update task status, save to IndexedDB outbox if offline[cite: 2].
SCR-05	Admin Rules & Triage Engine	Municipal Admin	Configure weights (\$W_m, W_c, W_v\$) and risk thresholds (\$R_c\$)[cite: 1, 2].	Parameter sliders, formula dry-run simulator, audit logs[cite: 2].	Adjust dynamic weights per season, set response escalation timers[cite: 1, 2].
SCR-06	Incident Audit & Analytics	Admin / Executive	Historical evaluation of response velocity and fuel savings[cite: 1, 2].	Data tables, CSV export toolbar, baseline response charts[cite: 2].	Filter immutable audit trails by city sector and export compliance reports[cite: 2].

Core Technology Stack[cite: 2]

Frontend Console[cite: 2]
Next.js 14, Tailwind CSS, Mapbox GL[cite: 2]

Ingestion Engine[cite: 2]
Python FastAPI on Google Cloud Run[cite: 1, 2]

Spatial Engine[cite: 2]
Uber H3 (PyH3 / H3-js Resolution 8)[cite: 1, 2]

AI & State Store[cite: 2]
Gemini 2.5 Flash & Google Firestore[cite: 1, 2]